

Checkpoint, 28 September 2026: the BPS uplift decoder of $\mathcal{N}=2$ SYK — verified structure, its variance, and deviation from freeness as nonzero-length wormholes

syk_gravity project

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Abstract

This note records where the project stands after the September 2026 audit and one day of follow-up work. It has two parts.

Part I restates the results about the uplift decoder $O = P_B(1 - n_i)P_B$ of one-flavor $\mathcal{N}=2$ SYK that survived the audit: exact structure, BPS dimensions, the Haar/Wachter *null* model, and robust numerics. It adds new checks: the charge-sector structure of the BPS space, and an exact index formula reproducing every measured BPS dimension. It also summarizes a new numerical campaign that measured the “chord invariants” (free cumulants, crossing weights, six-letter words) of the decoder for $q = 3$, $N' \leq 16$ and $q = 5$, $N' \leq 17$.

Part II is analytic.

- **The variance is a zero-energy two-point function.** The decoder variance is exactly the zero-energy two-point function of the neutral bilinear $n_i - \mu$ in a *single* R-charge sector.
- **The super-Schwarzian gets the structure but not the size.** We re-derive the super-Schwarzian (LMRS) answer from the zero-energy wavefunctions. It reproduces the parity and R-charge structure of the data with no free parameters, but overshoots the magnitude by 15–60 %. We trace this to the absence of a conformal window at $N' \lesssim 16$.
- **The finite- λ chord formula matches.** The double-scaled super-chord formula of Boruch, Lin and Yan [4], evaluated at $\lambda = 2p^2/N'$ with no free parameters, reproduces the measured variance to about 2 % for $N' = 5 \dots 16$.
- **Freeness corresponds to the zero-length wormhole.** Combining two results of [4] (our step), the BPS fraction, which is the Haar/Wachter value of the second free cumulant, is exactly the contribution of the *zero-length* supersymmetric wormhole within the double-scaled theory. The deviation from freeness is the contribution of wormholes of nonzero length:

$$r = a + \sum_{n \geq 1} P_n q^{2\Delta_n}.$$

Each statement is tagged as [\[literature\]](#), [\[derived\]](#), [\[numerical\]](#), [\[interpretation\]](#) or [\[conjecture\]](#).

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1 Scope, status labels, and notation

Status labels. [literature]: stated in a primary paper in papers/ (cited). [derived]: follows from stated inputs by the derivation shown, or checked here to machine precision. [numerical]: observation from exact numerics at the sizes available. [interpretation] / [conjecture]: reading or hypothesis, not established.

Model. One-flavor $\mathcal{N}=2$ SYK [2, 1]: N' complex fermions and a p -body supercharge $Q = \sum_{i_1 < \dots < i_p} C_{i_1 \dots i_p} \psi^{i_1} \dots \psi^{i_p}$ with i.i.d. complex Gaussian couplings, and $H = \{Q, Q^\dagger\}$. We use $p = q = \hat{q} = 3$ unless stated. In the code, Q acts as $\omega \wedge$ on $\Lambda^P(\mathbb{C}^{N'})$.

- N' is the size of the *enlarged* theory (old size $N = N' - 1$), and $P = \lfloor N'/2 \rfloor$.
- $D = \binom{N'}{P}$ is the dimension of the charge sector. B is the BPS subspace ($\ker Q \cap \ker Q^\dagger$) at charge P , $d = \dim B$, and P_B its projector.
- $a = d/D$ is the BPS fraction. $\tau = \text{Tr}_B / d$.

Decoder. $O = P_B \Pi P_B$ restricted to B , with $\Pi = 1 - n_i$ (the “slot” projector; $i = N' - 1$ is the uplift decoder of [1]). With i.i.d. couplings every mode is statistically equivalent, so we average over several modes.

- Eigenvalues $\lambda \in [0, 1]$ are squared cosines of principal angles between B and the slot.
- UV mean $b = \text{Tr}_D \Pi / D = 1 - P/N'$.
- Central quantity:

$$r \equiv \frac{\text{Var}(\lambda)}{b(1-b)}, \quad \text{Var}(\lambda) = \tau(O^2) - \tau(O)^2. \quad (1)$$

r is the fraction of the “UV” Bernoulli variance that survives BPS projection. $0 \leq r \leq 1$.

Part I — Verified structure and numerics

2 The safe core of the audit

The audit (`docs/project_archaeology.md`, `research/notes/gravitational_dual_fragility_audit.md`) checked the earlier draft claim by claim. What survived:

2.1 Exact structure

1. [literature] $0 \preceq O \preceq 1$, with $\lambda = \cos^2$ of the principal angles; definition as in [1, eqs. 101–102]. The BPS projector has the Hodge form $P_B = 1 - QGQ^\dagger - Q^\dagger GQ$, with G the Green’s function on the non-BPS part.
2. [derived] *Exact atoms.* $\lambda = 1$ eigenvectors are BPS states of the old theory in the “empty” copy that are annihilated by wedging with the new-mode couplings. $\lambda = 0$ eigenvectors are the analogous states in the “occupied” copy. (Characterization in the audit; implemented in `src/protected_atoms.py`.) [numerical] The atom fraction is 55.6% ($N = 8, 9$), 32.9% (10, 11), 5.6% (12, 13) and 0 (14–16). This is a monotone decay close to rank lower bounds, not a resonance.
3. [derived] *Mean and pairing.* The disorder-averaged mean is $\langle \lambda \rangle = b$ (exchangeability of modes plus $\sum_i n_i = P$). Per realization this holds only at even N' , half filling, where particle–hole symmetry pairs $\lambda \leftrightarrow 1 - \lambda$.
4. [derived] *Sum rule.* On the charge- P sector, $\sum_i (n_i - \mu) = 0$. Hence the BPS-projected covariance of two different modes is exactly $c = -1/(N' - 1)$. Measured: $-0.250, -0.200, \dots, -0.067$ for $N' = 5 \dots 15$.

2.2 BPS dimensions and charge sectors

1. [literature] $d = 2 \cdot 3^{N'/2-1}$ (even N' , $P = N'/2$) and $d = 3^{(N'-1)/2}$ (odd N') [1].
2. [derived] + verified. Fermion number fixes the R-charge. With ψ carrying charge $1/p$ and Q charge 1,

$$j = \frac{P - N'/2}{p} \quad \Rightarrow \quad j = 0 \text{ (} N' \text{ even)}, \quad j = -\frac{1}{6} \text{ (} N' \text{ odd)}. \quad (2)$$

So B is a *single* R-charge sector. Exact counts: even N' has BPS states only at $j = 0, \pm\frac{1}{3}$, with dimensions $(54; 27, 27)$, $(162; 81, 81)$, $(486; 243, 243)$ for $N' = 8, 10, 12$. The ratio $d_{\pm 1/3}/d_0 = \frac{1}{2}$ is exactly $\cos(\pi/3)/\cos 0$, as predicted by $\text{Tr } P_j \propto \cos \pi j$ [3, eq. 82]. Odd N' has $j = \pm\frac{1}{6}$.

3. [derived] + verified. The refined-index formula [4, eq. A.4],

$$\frac{d_j}{2^{N'}} = \frac{1}{p} \sum_{n=0}^{p-1} \sin^{N'}\left(\frac{\pi n}{p}\right) e^{i\pi j p} e^{-2\pi i j n}, \quad |j| < \frac{1}{2}, \quad (3)$$

reproduces *every* measured BPS dimension in every sector $|j| < \frac{1}{2}$ for $N' \leq 16$ (index saturation; test `tests/test_dssyk_n2.py`). The inverse Fourier transform resolves j only mod 1, hence the restriction.

2.3 The Haar / Wachter null model

Replace P_B by a Haar-random projector of the same rank. Free probability then predicts that the law of O is the *free compression* of $\text{Bern}(b)$ by a free projection of trace a : the Wachter (MANOVA) law.

- [literature] (Nica–Speicher [5]) Compressing a variable x by a free projection p of trace t gives

$$\kappa_n(pxp) = t^{n-1} \kappa_n(x) \quad (\text{free cumulants, computed in the compressed algebra}). \quad (4)$$

- Consequence [derived]: for the Haar null model, $\rho_n \equiv \kappa_n / \kappa_n^{\text{Bern}(b)} = a^{n-1}$. In particular

$$r_{\text{Haar}} = \rho_2 = a. \quad (5)$$

- [derived] (audit) Exact finite- D Weingarten moments hold, e.g. $m_2 = b[D^2(a+b-ab)-1]/(D^2-1)$, as does the genus-one resolvent $W_1(x) = -b(1-a)(1-b)x(x-1)/[(x-\lambda_-)(x-\lambda_+)]^{5/2}$ (checked through $k = 5$ and 6). These are properties of the *null model*. The audit found that the earlier “gravity = Weingarten genus expansion” reading is not supported.
- [numerical] Our Haar runs reproduce (5) and $\rho_{4,6} = a^{3,5}$ to three digits for $N' \geq 7$. This validates the numerical pipeline.

2.4 Robust numerics

[numerical] Interior level statistics of O are GUE ($\langle r_{\text{gap}} \rangle = 0.599 \pm 0.001$ at $N' = 12$ [1]). The interior density is *not* Wachter: mid-band weight is depleted about $2\times$ and moves to the walls. Of order 10–15% of the spectrum lies below the Wachter edge at $N' = 13$ –16.

2.5 Retracted

From the earlier draft (details in the audit):

- the end-of-world-brane dual with $e^{S_0} = D$;
- “BPS projection trivializes the Schwarzian”;
- “crossing = handle $\propto D^{-2}$ ”;
- the self-energy formula for O ;

- the “tail = image of the near-BPS spectrum” explanation;
- the bordered-Toeplitz closed form.

The Kunisky–Chebyshev error identity [6] holds for *any* pair of projections, so it is bookkeeping, not evidence.

3 Chord invariants of the decoder

What was measured. All quantities are exact linear algebra, per disorder realization (`src/chord_invariants.py`; `research/notes/chord_invariants_2026-09-28.md`). For centered operators $\hat{A}_i = P_B(\Pi_i - \mu)P_B$ with $v_i = \tau(\hat{A}_i^2)$:

- free cumulants κ_n ($n \leq 6$) of O , and the ratios $\rho_n = \kappa_n / \kappa_n^{\text{Bern}(b)}$;
- the crossing weight $X = \tau(\hat{A}_i \hat{A}_j \hat{A}_i \hat{A}_j) / (v_i v_j)$ and the non-crossing factorization $F = \tau(\hat{A}_i \hat{A}_i \hat{A}_j \hat{A}_j) / (v_i v_j)$ for $i \neq j$;
- all six-letter pair words, compared with X^{cr} , where cr is the crossing number.

The fully crossing word $ijkijk$ discriminates between two rules: free compression predicts X^2 and a single-parameter chord (q -Gaussian) rule predicts X^3 . We quote $e_3 = \ln w_3 / \ln X$.

q	N'	model	a	r	ρ_4 (a^3)	X	F	e_3
3	10	SYK	0.643	0.746	0.435 (0.266)	0.810	0.995	2.29 ± 0.01
3	12	SYK	0.526	0.687	0.348 (0.146)	0.785	0.995	2.35 ± 0.01
3	14	SYK	0.425	0.642	0.290 (0.077)	0.769	0.994	2.41 ± 0.01
3	16	SYK	0.340	0.601	0.245 (0.039)	0.755	0.995	2.42 ± 0.03
3	16	Haar	0.340	0.340	0.039	0.346	1.003	1.94
5	17	SYK	0.874	0.908	0.751 (0.668)	0.931	0.998	$w_3/X^2 = 0.979$
5	17	Haar	0.874	0.874	0.668	0.874	1.000	$w_3/X^2 = 1.000$

Table 1: [numerical] Selected chord invariants (mean over realizations; 60→2 realizations per point as N' grows; full tables in `results/data/chord_invariants_2026-09-28/`).

Findings [numerical].

1. **SYK is not free compression, and the gap grows with N' .** r/a rises from 1.04 ($N' = 7$) to 1.77 ($N' = 16$).
2. **The single-operator law is “Wachter with a renormalized parameter” only in its low cumulants.** $\rho_n^{1/(n-1)}$ nearly coincide, but the KS distance of the spectrum to $\text{Wachter}(t = r)$ is 0.15–0.23.
3. **Words with up to two crossings are multiplicative.** $F = 1$ and $\text{cr} \leq 2$ words equal X^{cr} to $\lesssim 1\%$.
4. **The fully crossing word obeys neither rule.** $e_3 \simeq 2.3$ –2.4, drifting slowly upward: neither free (2) nor single-parameter chord (3). At $q = 5$, $X \approx 0.95$ makes e_3 ill-conditioned, but the ratios show the same pattern.
5. **Rigidity grows but decelerates.** $\eta_r = (r - a)/(1 - a)$ grows as 0.29, 0.34, 0.38, 0.40 ($N' = 10$ –16). Section 7 explains this.

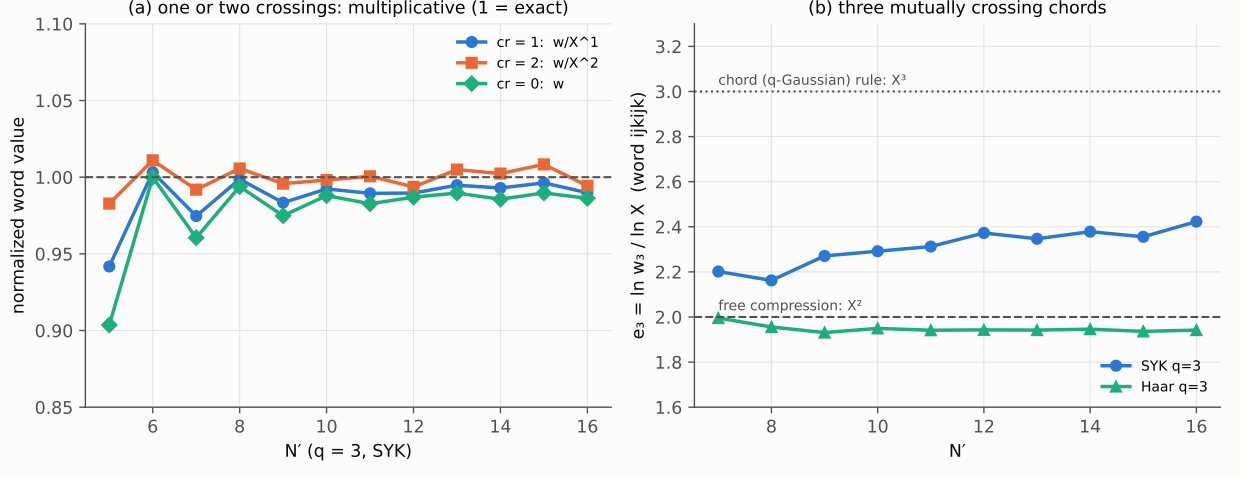


Figure 1: [numerical] Six-letter pair words, $q = 3$ SYK. (a) Words with 0, 1, 2 crossings are multiplicative in X . (b) Effective exponent of the fully crossing word: Haar follows free compression ($e_3 \approx 2$). SYK lies between the free and single-parameter chord rules. Error bars (≈ 0.01) are smaller than the markers.

Part II — The decoder variance and the wormhole

4 The variance is a zero-energy two-point function

Proposition 1 ([derived], exact). *Let $\mu = \tau(P_B n_i P_B)$ and $\mathcal{O} = n_i - \mu$. Then*

$$\text{Var}(\lambda) = \frac{\text{Tr}[P_j \mathcal{O} P_j \mathcal{O}]}{\text{Tr} P_j}, \quad P_j \equiv P_B = \lim_{\beta \rightarrow \infty} e^{-\beta H} \Big|_{\text{charge } P}, \quad (6)$$

with j given by (2).

Proof. Var is invariant under $\Pi \rightarrow 1 - \Pi$, so we may use n_i . Then $\tau((P_B n P_B)^2) - \tau(P_B n P_B)^2 = \tau(P_B n P_B n P_B) - \mu^2 = \tau(P_B \mathcal{O} P_B \mathcal{O} P_B)$, since $P_B^2 = P_B$ and $\tau(P_B \mathcal{O} P_B) = 0$. Fermion number is conserved, so P_B is the zero-energy projector of the single R-charge sector (2). $\square$

So $\text{Var}(\lambda)$ is the “infinitely-long-time” two-point function of $\mathcal{O}$ in the sense of LMRS [3] (their $\hat{\mathcal{O}} = POP$), with two differences:

- it is normalized by the number of BPS states in *one* sector, $\text{Tr} P_j$, not the total N_{BPS} ;
- $\mathcal{O}$ is the *same-site* bilinear $\bar{\psi}_i \psi_i$, not LMRS’s $\psi^k \bar{\psi}_i$ with $i \neq k$.

At leading large N the two bilinears have the same conformal two-point function, $\langle n_i(t) n_i(0) \rangle_c = \langle \bar{\psi}_i(t) \psi_i(0) \rangle \langle \psi_i(t) \bar{\psi}_i(0) \rangle + O(1/N)$. This is a single cross-contraction, so both have $\Delta = 2 \cdot \frac{1}{2q} = \frac{1}{3}$.

[numerical] In our BPS sectors, $\text{Var}_{\text{hop}} / \text{Var}_{n_i} = 1.14, \dots, 1.06$ for $N' = 8 \dots 14$, tending to 1.

5 The super-Schwarzian answer, and why it overshoots

5.1 Derivation from the zero-energy wavefunctions

Imported [literature] from LMRS [3].

- The two-sided zero-energy state $|Z_j\rangle$ of $\mathcal{N}=2$ super-Liouville quantum mechanics has two fermionic components, $g_{1,2} \propto e^{-\ell/2} K_{\frac{1}{2} \mp j}(2e^{-\ell/2})$ (their eq. 22), normalizable for $|j| < \frac{1}{2}$.
- A neutral matter operator of dimension Δ acts on it as multiplication by $e^{-\Delta\ell}$ (their eq. 54).
- The $\beta \rightarrow \infty$ two-sided state in sector j is $A_{z,j}|Z_j\rangle$ (their eqs. 37, 39).

Step 1 [derived]: reduction to a normalized expectation value. $\text{Tr}[P_j \hat{O} P_j \hat{O}] = |A_{z,j}|^2 \langle Z_j | e^{-\Delta\ell} | Z_j \rangle$. At $\Delta = 0$ this must equal $\text{Tr} P_j$, so

$$\frac{\text{Tr}[P_j \hat{O} P_j \hat{O}]}{\text{Tr} P_j} = \frac{\langle Z_j | e^{-\Delta\ell} | Z_j \rangle}{\langle Z_j | Z_j \rangle}. \quad (7)$$

Neither e^{S_0} nor the TFD coefficient enters; only the shape of the wavefunction does.

Step 2 [derived]: a Mellin integral. For a neutral operator the $U(1)$ integral is trivial. With $x = 2e^{-\ell/2}$ ($d\ell = -2dx/x$, $e^{-\ell} = x^2/4$, $e^{-\Delta\ell} = (x/2)^{2\Delta}$),

$$I_\nu(\Delta) \equiv \int_{-\infty}^{\infty} d\ell e^{-(1+\Delta)\ell} K_\nu(2e^{-\ell/2})^2 = 2^{-1-2\Delta} \int_0^\infty x^{1+2\Delta} K_\nu(x)^2 dx.$$

Use the Mellin transform (Gradshteyn–Ryzhik 6.576.4 with equal arguments [7]; checked numerically to 10^{-11}):

$$\int_0^\infty x^{s-1} K_\nu(x)^2 dx = \frac{\sqrt{\pi} \Gamma(\frac{s}{2}) \Gamma(\frac{s}{2} - \nu) \Gamma(\frac{s}{2} + \nu)}{4 \Gamma(\frac{s+1}{2})}, \quad \text{Re } s > 2|\nu|.$$

Set $s = 2 + 2\Delta$ and sum the two components $\nu = \frac{1}{2} \mp j$. Using $\Gamma(z+1) = z\Gamma(z)$ on the second factors,

$$\Gamma(\Delta + \frac{1}{2} + j) \Gamma(\Delta + \frac{3}{2} - j) + \Gamma(\Delta + \frac{1}{2} - j) \Gamma(\Delta + \frac{3}{2} + j) = (2\Delta + 1) \Gamma(\Delta + \frac{1}{2} + j) \Gamma(\Delta + \frac{1}{2} - j).$$

Then $(2\Delta + 1)/\Gamma(\Delta + \frac{3}{2}) = 2/\Gamma(\Delta + \frac{1}{2})$, and Legendre duplication, $\Gamma(\Delta) \Gamma(\Delta + \frac{1}{2}) = 2^{1-2\Delta} \sqrt{\pi} \Gamma(2\Delta)$, give

$$\mathcal{N}_j(\Delta) \equiv I_{\frac{1}{2}-j} + I_{\frac{1}{2}+j} = \frac{1}{8} \frac{\Delta \Gamma(\Delta)^2 \Gamma(\Delta + \frac{1}{2} + j) \Gamma(\Delta + \frac{1}{2} - j)}{\Gamma(2\Delta)}. \quad (8)$$

The convergence condition at $\Delta = 0$ is exactly $|j| < \frac{1}{2}$, the normalizability condition of $|Z_j\rangle$.

Step 3 [derived]: normalize. $\mathcal{N}_j(0) = \frac{1}{4} \Gamma(\frac{1}{2} + j) \Gamma(\frac{1}{2} - j) = \pi/(4 \cos \pi j)$. This reproduces the norm $\langle Z_j | Z_j \rangle \propto 1/\cos \pi j$ of [3, eq. 26], a check on Step 2. Hence

$$\langle e^{-\Delta\ell} \rangle_j = \frac{\cos \pi j}{2\pi} \frac{\Delta \Gamma(\Delta)^2 \Gamma(\Delta + \frac{1}{2} \pm j)}{\Gamma(2\Delta)} \equiv F(\Delta, j), \quad (9)$$

which is exactly LMRS eq. (66) divided by $\text{Tr} P_j = e^{S_0} \cos \pi j$.

Step 4 [literature]: SYK normalization. Converting from Schwarzian to SYK units multiplies by $b_q^2(2CJ)^{-2\Delta}$, with $b_q = [\tan(\pi/2\hat{q})/2\pi]^{1/\hat{q}}$ and $C = \alpha_S N/J$, $\alpha_S = 0.00842$ for $\hat{q} = 3$ (LMRS eqs. 64, 78–80). Therefore

$$r_{\text{LMRS}}(N') = \frac{b_q^2 (2\alpha_S N')^{-2/3} F(\frac{1}{3}, j)}{b(1-b)} \quad (10)$$

with no free parameter.

- Checks: the code reproduces the LMRS Table 1 Schwarzian column (0.1112 vs 0.111; 0.0874 vs 0.0874); $F(0, j) = 1$; J cancels.
- Note: LMRS eq. (77) prints $\langle C\bar{C} \rangle = 2J/N$. FGMS [2, eq. 5.4] has $2J/N^2$, which is required for extensivity. This matters only for gap-based checks.

5.2 Comparison

N'	9	10	11	12	13	14	15	16
r (SYK)	0.728	0.746	0.666	0.687	0.617	0.642	0.577	0.601
r_{LMRS}	0.957	0.957	0.833	0.847	0.744	0.765	0.675	0.700
Haar $r = a$	0.643	0.643	0.526	0.526	0.425	0.425	0.340	0.340

Table 2: [numerical] Decoder variance ratio against the parameter-free super-Schwarzian prediction (10).

What works [numerical].

- **Parity.** The j -factor $F(\frac{1}{3}, -\frac{1}{6})/F(\frac{1}{3}, 0) = \cos \frac{\pi}{6} \Gamma(1)\Gamma(\frac{2}{3})/\Gamma(\frac{5}{6})^2 = 0.920$ removes the even/odd alternation of r with no parameters.
- **A pre-registered test** at fixed even N' , comparing the $j = \pm \frac{1}{3}$ sectors ($P = N'/2 \pm 1$) with $j = 0$. Predicted $\text{Var}_{1/3} / \text{Var}_0 = \frac{1}{2} \Gamma(\frac{7}{6})\Gamma(\frac{1}{2})/\Gamma(\frac{5}{6})^2 = 0.645$. Measured: 0.642, 0.668, 0.677, 0.683 ($N' = 8 \dots 14$). The Haar value is $0.586 \rightarrow 0.560$, and “no j -dependence” gives 1. So this is a 6% success, although the measured ratio drifts away from the prediction.

What fails [numerical]. The magnitude is too high by 15–60%. The N' -dependence also does not match: the pure $N'^{-2/3}$ shape with free normalization gives $\chi^2/\text{dof} = 335$. A fitted $1 - 2.2/N'$ factor restores agreement, but so does a plain power $N'^{-0.44}$. That fitted factor is not a prediction.

5.3 Diagnosis: no conformal window [derived]

By Steps 1–4, the prediction is the zero-energy average of the conformal matter correlator $b_q^2(Jt)^{-2\Delta}$ with $Jt(\ell) = 2\alpha_S N' e^{\ell/2}$. The weight $\rho_j(\ell) e^{-\Delta\ell}$ fixes which physical times dominate:

N'	10	15	20	50	120
median Jt	0.50	0.75	1.0	2.5	6.0
weight where conformal value > UV maximum $\frac{1}{4}$	67%	49%	35%	4%	0

At our sizes the “zero-energy” correlator is dominated by times *shorter than* $1/J$. There the conformal input overshoots the true correlator, which is bounded by its equal-time value. The effective small parameter is $1/(2\alpha_S N') \approx 60/N'$, and a clean window needs $N' \sim 10^2$. [numerical] Consistent with this: the BPS gap is $0.19J$ at $N' = 14$ (not $\ll J$), and $r_{\text{LMRS}} > 1$ for $N' \leq 8$. A parameter-free toy that caps the conformal correlator at its UV value brings the prediction within 0–10% of the data, for two different operators.

6 The finite- λ super-chord answer

The double-scaled theory ($N, p \rightarrow \infty$ with $\lambda = 2p^2/N$ fixed, $q = e^{-\lambda}$) is solvable by chord methods and is UV-complete. The super-Schwarzian is its triple-scaling limit $\lambda \rightarrow 0$ [4]. Boruch, Lin and Yan construct the two-sided super-chord Hilbert space (chord number n = discrete wormhole length, $\ell = 2\lambda n$) and its zero-energy Hartle–Hawking state $|\Psi, j\rangle$ ([literature], their eqs. 3.1–3.11). They compute the normalized zero-temperature two-point function of a neutral operator ([literature], their eqs. 4.5–4.8):

$$\frac{\langle \Psi, j | q^{2\Delta n} | \Psi, j \rangle}{\langle \Psi, j | \Psi, j \rangle} = \frac{(q^{2+4\Delta}; q^2)_\infty (q^{1\pm 2j}; q^2)_\infty (q^2; q^2)_\infty}{(q^{2+2\Delta}; q^2)_\infty^2 (q^{1+2\Delta\pm 2j}; q^2)_\infty}. \quad (11)$$

As $\lambda \rightarrow 0$ this tends to $(2\lambda)^{2\Delta} F(\Delta, j)$, i.e. to (9) (checked: ratio $1 - 3 \times 10^{-3}$ at $\lambda = 0.3$).

Identification with r [derived]. BLY normalize the matter operator so that $W_L W_R \propto q^{\Delta(n_O + n_X)}$ with unit constant, i.e. its infinite-temperature two-point function in the charge sector equals 1. Since $|\Psi, j\rangle \propto \Pi_{0,j} |\Omega, j\rangle$, eq. (11) equals $[\text{Tr}(P_j W P_j W) / \text{Tr} P_j] / [\text{Tr}_j(WW) / \text{Tr}_j \mathbb{K}]$. For $W = n_i - \mu$ on Λ^P , the denominator is exactly $b(1 - b)$. By Proposition 1,

$$r = \text{eq. (11)} \Big|_{\lambda=2p^2/N', \Delta=1/p, j=(P-N'/2)/p} \quad (12)$$

Here $\Delta = 1/p$ because $\bar{\psi}_i \psi_i$ has one ψ and one $\bar{\psi}$ (BLY eq. 1.8). No conformal normalization and no α_S enter.

N'	j	λ	r (SYK)	r chord	r_{LMRS}	SYK/chord	a exact	a chord
6	0	3.00	0.905	0.913	1.345	0.990	0.900	0.900
8	0	2.25	0.819	0.832	1.110	0.984	0.771	0.789
10	0	1.80	0.746	0.756	0.957	0.987	0.643	0.671
12	0	1.50	0.687	0.690	0.847	0.996	0.526	0.559
13	$-\frac{1}{6}$	1.38	0.617	0.613	0.744	1.007	0.425	0.456
14	0	1.29	0.642	0.635	0.765	1.011	0.425	0.459
15	$-\frac{1}{6}$	1.20	0.577	0.565	0.675	1.021	0.340	0.371
16	0	1.13	0.601 ± 0.005	0.589	0.700	1.022	0.340	0.373
14	$-\frac{1}{3}$	1.29	0.449	0.418	0.504	1.074	0.243	0.265

Table 3: [numerical] Exact $q = 3$ SYK against the parameter-free finite- λ prediction (12). The BPS fraction is from the exact index (3) and from the double-scaled formula $D(j)$ [4, eq. 3.17].

Results [numerical].

- r agrees within 2.2% for every $N' = 5 \dots 16$, including parity, with no rescaling (Fig. 2). The super-Schwarzian misses by 15–60% over the same range.

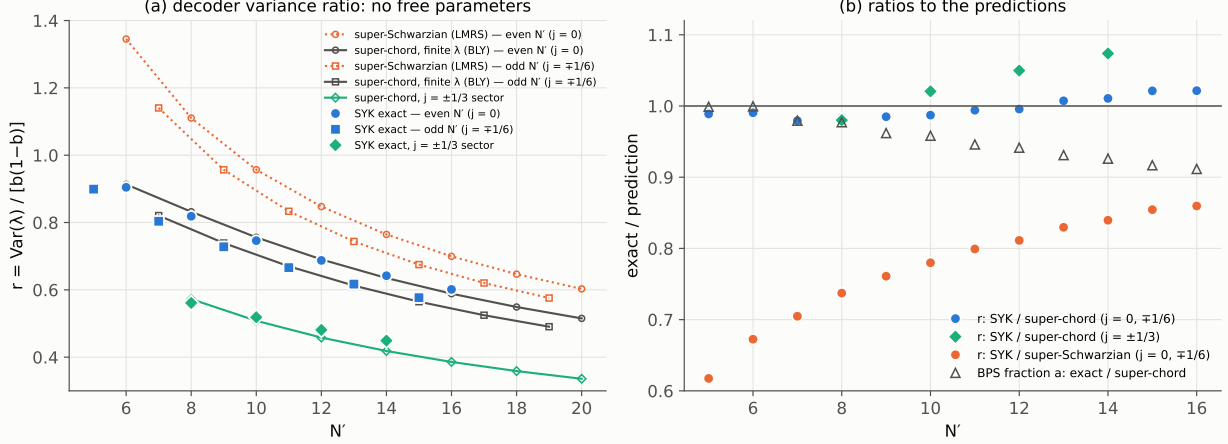


Figure 2: [numerical] (a) r versus N' : exact SYK (filled), finite- λ super-chord (solid; eq. 12), super-Schwarzian (dotted; eq. 10); no free parameters. (b) Ratios to the predictions, including the BPS fraction a .

- There is a small systematic drift in SYK/chord ($0.98 \rightarrow 1.02$). In the $j = \pm \frac{1}{3}$ sectors the agreement degrades to +7% at $N' = 14$.
- The double-scaled BPS fraction overshoots the exact one by up to 10%. This is a pure finite- p effect: $\cos^{N'}(\pi/2p)$ versus $e^{-N'\pi^2/8p^2}$.

Caveats.

- The λ identification is BLY's standard $\lambda = 2p^2/N$, fixed before comparing. The prediction scales as $r \propto \lambda^{0.4-0.6}$, so a 5% ambiguity in λ moves r by about 2.5%. The 2% agreement is at the resolution of the double-scaled mapping at $p = 3$.
- The double-scaled theory at $\lambda = 18/N'$ is not the $N' \rightarrow \infty$ limit at fixed $p = 3$. Its Schwarzian coupling is $\alpha_S = 1/(8p^2) = 0.0139$ versus 0.00842. Agreement must eventually degrade at large N' ; the positive drift may be its onset.

Interpretation [interpretation]. Together with the diagnosis above: at accessible sizes the decoder variance sits in a *double-scaled* regime ($\lambda \approx 1.1-3.6$), not a Schwarzian one. The chord theory, as the UV completion, captures the crossover the Schwarzian misses. BLY draw the same lesson for the ground-state degeneracy itself.

7 Deviation from freeness = wormholes of nonzero length

The project's framing asks for “the deviation from freeness” of the decoder law. The chord theory gives this deviation a geometric meaning, at least for the second free cumulant. Because this section mixes results of BLY with steps of our own, we first record the provenance of each ingredient.

7.1 Provenance

Ingredient	Source
BPS fraction = probability that the SUSY wormhole has zero chord number: $D(j) = P_0$	[literature] BLY eq. 1.1, derived in eqs. 3.12–3.15, closed form eq. 3.17
Zero-temperature two-point function = norm sum (3.10) with $q^{2\Delta_n}$ inserted; each term over the norm (3.11) is the length probability P_n	[literature] BLY eqs. 4.5–4.6 and the sentence after eq. 3.11; closed form eqs. 4.7–4.8
That normalized two-point function equals our r	[derived] here: BLY’s operator normalization + Proposition 1 (eq. 12)
Splitting off $n = 0$; $\eta_r = \langle q^{2\Delta_n} \rangle_{n \geq 1}$	[derived] here (algebra, using $\sum_{n \geq 1} P_n = 1 - P_0$)
Treatment of the $n = 0$ term	[derived] here + numerical check (see below); not spelled out by BLY
Identifying P_0 with the Haar/Wachter value of ρ_2	[interpretation] here. BLY do not discuss freeness. The Haar value $\rho_2 = a$ comes from free compression, eq. (5)

Notation. Write $a_\lambda \equiv D(j)$ for the double-scaled BPS fraction (BLY eq. 3.17), to distinguish it from the exact SYK fraction $a = d/D$ of eq. (3). At $p = 3$ they differ by 0–10 % (Table 3; $a_\lambda > a$, a finite- p effect). Likewise r_λ denotes eq. (11).

7.2 Two facts about the same distribution

Let P_n be the probability that the supersymmetric wormhole $|\Psi, j\rangle$ has chord number n : the n -sector weight of $|\Psi, j\rangle$ divided by its norm (BLY eq. 3.11). States of different n are orthogonal.

Fact 1 [literature] [4, eqs. 1.1, 3.12–3.15]. Slice the fixed-charge partition function two-sidedly: $Z(j, \beta)/Z(j, 0) = \langle \Omega, j | e^{-\beta H} | \Omega, j \rangle / \langle \Omega, j | \Omega, j \rangle$, where $|\Omega, j\rangle$ is the empty ($n = 0$) wormhole. As $\beta \rightarrow \infty$, $e^{-\beta H}$ projects onto the chord ground state:

$$a_\lambda = \frac{Z(j, \infty)}{Z(j, 0)} = \frac{|\langle \Omega, j | \Psi, j \rangle|^2}{\langle \Omega, j | \Omega, j \rangle \langle \Psi, j | \Psi, j \rangle} = P_0. \quad (13)$$

Fact 2 [literature] + [derived]. BLY write the normalized zero-temperature two-point function as the norm sum with $q^{2\Delta_n}$ inserted (their eqs. 4.5–4.6), i.e. as $\sum_n P_n q^{2\Delta_n}$; $W_L W_R = q^{2\Delta_n}$ is a scalar on each n -sector. By our identification (12),

$$r_\lambda = \sum_{n \geq 0} P_n q^{2\Delta_n}. \quad (14)$$

The $n = 0$ term [derived]. As written, the $n = 0$ summand of BLY’s sums (3.10) and (4.6) is not well defined: its prefactor $(q^2; q^2)_{-1}$ is singular. We take the $n \geq 1$ terms from their formula and set $P_0 = a_\lambda$ from Fact 1. This is not circular. With that assignment, *both* of BLY’s independent closed forms are reproduced:

$$\sum_{n \geq 0} P_n = 1 \quad (\text{norm, eq. 3.11}), \quad \sum_{n \geq 0} P_n q^{2\Delta_n} = r_\lambda \quad (\text{eq. 4.7}),$$

to 10^{-10} or better for $N' = 8, 12, 14, 16$ (hh_length_distribution, test test_free_compression_is_zero_length_term). BLY do not spell out this bookkeeping; the check is ours.

7.3 The identity

Since $q^{2\Delta_0} = 1$, Facts 1–2 combine into ([derived], our step)

$$r_\lambda = \underbrace{a_\lambda}_{=P_0} + \sum_{n \geq 1} P_n q^{2\Delta_n}, \quad \eta_r^{(\lambda)} \equiv \frac{r_\lambda - a_\lambda}{1 - a_\lambda} = \langle q^{2\Delta_n} \rangle_{n \geq 1} \quad (15)$$

Equation (15) is an exact statement *within the double-scaled theory*. The algebra is trivial; the content is in Facts 1–2 (BLY) and in the identification of r_λ with our r (Sec. 6).

Relation to freeness [interpretation]. In the Haar null model the second free cumulant ratio is $\rho_2 = a$ (eq. 5; free compression, Nica–Speicher). In the chord theory the zero-length term is a_λ , which is the chord theory’s own value of the BPS fraction. So, *within the double-scaled theory*, the zero-length wormhole contributes exactly the value that free compression would assign to r at that BPS fraction. We read this as:

- the Haar/Wachter value of the second free cumulant corresponds to the zero-length wormhole;
- the deviation from freeness, $r - a$, corresponds to wormholes of nonzero length, each weighted by the matter propagator $q^{2\Delta_n} = e^{-\Delta_\ell}$;
- the rigidity fraction η_r is that propagator averaged over nonzero-length wormholes.

Three caveats limit this reading:

1. *Equality of values, not of structures.* Both sides equal the BPS fraction. We have not shown that free compression and the zero-length wormhole are the same mechanism. Only the second cumulant is tested; the extension to the full law is Conjecture 3 below.
2. *a_λ versus a .* The exact SYK statement uses a ; the chord identity uses a_λ , which is 0–10 % larger at $p = 3$. The two coincide only within the double-scaled approximation. This is why the measured $\eta_r = (r - a)/(1 - a)$ in Fig. 3(b) sits about 0.04 above the chord curve built from a_λ .
3. *$r \geq a$ is automatic in the chord theory, not an exact finite- N theorem.* [numerical] The realization-averaged r satisfies it within errors at every (q, N') measured, and by many standard errors for $q = 3$, $N' \geq 7$ (e.g. $r/a = 1.77$ at $N' = 16$). Individual realizations violate it only where $a \gtrsim 0.9$ and d is tiny ($q = 3$, $N' = 5, 6$; $q = 5$, $N' = 9, 10$; 2 of 300 at $q = 3$, $N' = 7$). There $r \approx a$ and sample fluctuations cross; the minimum single-realization r/a is 0.965.

Figure 3(a) shows the decomposition at $N' = 14$: $a_\lambda = P_0 = 0.459$ of $r_\lambda = 0.635$ comes from $n = 0$, and the remaining 0.176 from $n = 1, 2, 3, \dots$ (For comparison, the exact values are $a = 0.425$ and $r = 0.642$.)

7.4 Consequences

1. **η_r turns over [numerical]/[interpretation].** In the chord theory η_r *peaks* near $N' \approx 22$ ($\eta_r \approx 0.36$) and then decays. The measured deceleration ($0.34 \rightarrow 0.38 \rightarrow 0.40$ at $N' = 12, 14, 16$) is the approach to this maximum, not a plateau. This is a sharp, testable prediction for $N' \gtrsim 20$ (stochastic trace estimation). The turnover is robust beyond the double-scaled approximation: at fixed $p = 3$, LMRS also gives $r \propto N'^{-2/3} \rightarrow 0$.

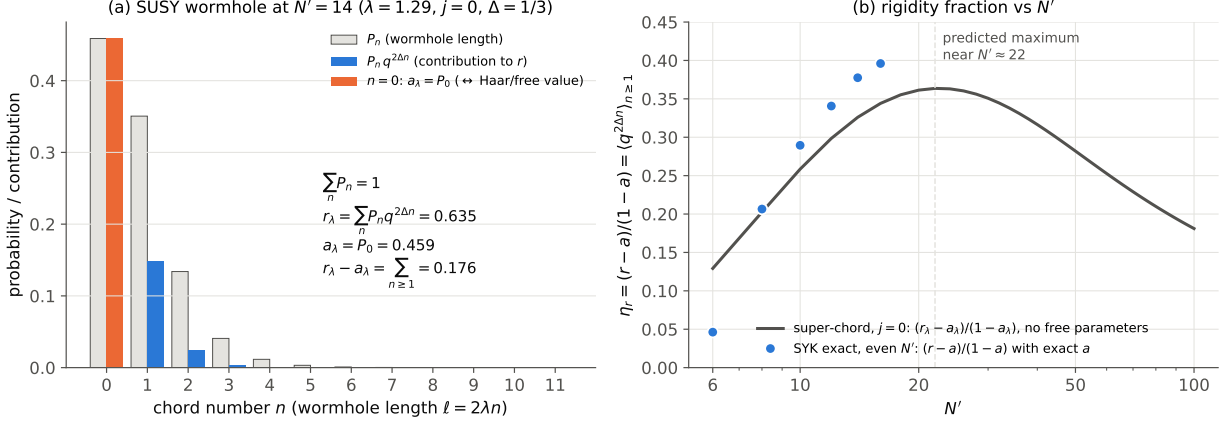


Figure 3: (a) [derived] Wormhole-length distribution P_n of the supersymmetric Hartle–Hawking state at $N' = 14$ ($\lambda = 18/14, j = 0$) and its contribution $P_n q^{2\Delta n}$ to r . The $n = 0$ bar is $P_0 = a_\lambda$, the double-scaled BPS fraction, which corresponds to the Haar/free value (Sec. 7.1). (b) [numerical] vs prediction: η_r from the chord theory (line; no parameters) and exact SYK (points). The chord curve peaks near $N' \approx 22$. The measured values lie about 0.04 above it, consistent with the finite- p excess of a_{chord} over the exact a . At $N' = 6$, $1 - a = 0.1$, so η_r is poorly conditioned.

2. **Q1 settled structurally.** $a \sim e^{-\pi^2 N'/72}$ in the chord theory (exactly, at $p = 3, a \propto \cos^{N'}(\pi/6)$ up to the binomial normalization) is exponentially small, while $r \sim \lambda^{2\Delta} \propto N'^{-2/3}$ is power-law small. So $r/a \rightarrow \infty$: the decoder law moves ever further from Wachter, while $r \rightarrow 0$ (ever further from the commuting UV).
3. **Conjecture [conjecture].** The *entire* free-compression (Wachter/MANOVA) law is the zero-length truncation of the super-chord computation of the decoder moments. Only m_2 has been tested.

8 Open problems and next steps

1. **Zero-temperature OTOC: X and e_3 .** BLY §5 constructs one-particle wormholes and the operator $W_L W_R \sim q^{\Delta(n_O + n_X)_{\text{tot}}}$ (their eq. 5.18), but does not evaluate the zero-temperature OTOC. For our operators the matter–matter crossing weight must be site-local: 1 for disjoint sites, since those bilinears commute, rather than the random-operator q^{Δ^2} . This gives a parameter-free prediction of $X \approx 0.73\text{--}0.85$ and $e_3 \approx 2.3\text{--}2.4$ (Table 1).
2. **Higher moments and the density.** $m_k = \tau(O^k)$ are zero-temperature k -point functions of *one fixed projector*. Writing $n_i = \bar{\psi}_i \psi_i$ with ψ_i a single-fermion chiral chord restores a Wick structure, because the infinite-temperature state is Gaussian. Same-site fermion chords then cross with a sign. Testing the conjecture above requires these.
3. **Finite- p corrections.** Explain the $\pm 2\%$ drift and the $j = \pm \frac{1}{3}$ deviation. Candidates: exact finite- N intersection weights (e.g. $1 - 2p/N$ vs $e^{-2p/N}$ for a single-site bilinear crossing a 3-body chord), and the λ identification.
4. **Larger N' .** Test the η_r maximum near $N' \approx 22$ and the breakdown of double scaling.
5. **Framing.** The project overview’s “planar chords = non-crossing partitions (Narayana)

$\Rightarrow$ Wachter” does not describe the actual $\mathcal{N}=2$ chord rules. Those have directed chords, friend/enemy weights and q^Δ matter weights, and give Wachter only as the zero-length piece.

A Reproducibility

All numbers are exact linear algebra (no stochastic trace estimation), with the same coupling generator and seeds as `src/decoder_fast.py`. Code state: commit 081877d plus the working-tree files added for this note (`src/dssyk_n2.py` [hh_length_distribution], `scripts/plot_wormhole_freeness.py`).

Item	Code / data
Chord invariants, Tab. 1, Fig. 1	<code>scripts/run_chord_invariants.py</code> , <code>scripts/chord_invariants_campaign.sh</code> (groups A–F2), <code>scripts/analyze_chord_invariants.py</code> ; <code>results/data/chord_invariants_2026-09-28/</code> (JSONL per realization, <code>meta_*.json</code> with args and commit)
Super-Schwarzian, Tab. 2	<code>src/lmrs_predictions.py</code> , <code>scripts/compare_lmrs_variance.py</code> , <code>scripts/lmrs_mismatch_checks.py</code> , <code>scripts/lmrs_jsector_test.py</code> ; <code>results/data/lmrs_variance_2026-09-28/</code>
Super-chord, Tab. 3, Figs. 2–3	<code>src/dssyk_n2.py</code> , <code>scripts/compare_dssyk.py</code> , <code>scripts/plot_wormhole_freeness.py</code> ; <code>results/data/dssyk_2026-09-28/</code>
Tests	<code>tests/test_chord_invariants.py</code> , <code>tests/test_lmrs_predictions.py</code> , <code>tests/test_dssyk_n2.py</code>
Derivations (living)	<code>docs/derivations.md</code> D1, D2

Realizations. $q = 3$: 60/30 (SYK/Haar) at $N' = 5$ –8; 40/20 at 9–10; 24/12 at 11; 16/8 at 12; 10/5 at 13; 4/2 at 14; 3/2 at 15; 2/1 at 16. $q = 5$: 20/10 at 9–14; 8/4 at 15; 4/2 at 16; 2/1 at 17. j -sector test: seeds 0–3, $N' = 8$ –14.

Memory safety. The two largest runs ($q = 3$, $N' = 16$; $q = 5$, $N' = 17$) were OOM-killed once when run concurrently. They were rerun after two changes: a partial eigendecomposition of $\{Q, Q^\dagger\}$ with a certified spectral gap, and a byte-bounded cache. Regression against stored rows: $\leq 4 \times 10^{-16}$. All heavy jobs now run under `scripts/memwatch.py` (per-job cap, 30 GB project budget). Peaks were 7.5 and 7.0 GB.

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